

Platonic 3D Folding, Octad Triad Closures and Modified Spiral Dynamics in the Golay-Leech UBP Substrate

Euan R A Craig
Auckland, New Zealand

20 March 2026

Abstract

This study examines geometric phenomena emergent from the UBP substrate implemented in UBP Core Studio v4.0 (UBP Core v5.7). Using the extended binary Golay code (24, 12, 8) as the underlying 4096-state universe, augmented by the Leech-based symmetry tax and UBP-Py dynamics, I probe for platonic, folding, and relational structures.

I find three core insights: (i) the 759 octads induce a perfectly regular graph of degree 280 with 35,420 XOR-closed triads and near-strongly-regular common-neighbor counts; (ii) a canonical 3-fold pairwise-XOR compression maps the entire codebook to a balanced 4-symbol core-tension alphabet {000, 011, 101, 110}; and (iii) the original UBP-Py spiral operator is an involution (2048 disjoint 2-cycles), while a 1-bit spatial shift plus MATH_CONST_PHI_001 XOR injection unlocks 1953-step orbits partitioning 3906 states into two symmetric hemispheres of growth, with Origin (Zero) and Hydrogen vectors locked into these cycles.

All results are fully reproducible from the GitHub repository (scripts, CSV/JSON, and figures). I validate against internal UBP laws (marked as 'LAW_EXAMPLE_001'), discuss platonic interpretations, and note next steps that may be taken towards formal classification and canonical operators.

The Universal Binary Principal is an experimental system I am developing so caution is advised - I attempt pure transparency and accuracy but check results for yourself.

1 Introduction

The UBP substrate uses the extended binary Golay code, Leech-lattice symmetry taxation, and deterministic Python VM dynamics (UBP-Py) in a computational geometry engine. The primary objective of this investigation - conducted entirely within UBP Core Studio v4.0 from the public repository https://github.com/DigitalEuan/UBP_Repo/tree/main/core_studio_v4.0 was to determine whether these primitives spontaneously generate platonic-solid-like, folding, or symmetry-tax phenomena interpretable as deeper structural laws.

This paper consolidates the raw investigatory study (detailed in `findings.md`) with the subsequent audit (`ubp_audit_1.txt`).

2 UBP Primitives

2.1 Symmetry Tax

The Leech engine defines the tax on a 24-vector v as

$$\text{Tax}(v) = (\text{HammingWeight}(v) \times Y) + \frac{\|v\|^2}{8}.$$

For binary (0/1) Golay codewords, $\|v\|^2 = \text{HammingWeight}(v)$, collapsing the expression to

$$\text{Tax}(\text{codeword}) = w \times \left(Y + \frac{1}{8} \right),$$

where w is the weight. Thus, within the strict Golay codebook, tax depends only on weight. Non-trivial tax geometry must therefore arise from relational structures (triads) or excursions outside the binary set. Full definition and implementation appear in the main UBP system repository documentation https://github.com/DigitalEuan/UBP_Repo/blob/main/core_studio_v4.0/core/ubp_files_and_usage.md.

2.2 Folding Operator

A natural recursive pairwise-XOR fold consistent with LAW_GEO_FOLD_001 is applied:

$$v'[i] = v[2i] \oplus v[2i + 1].$$

Three successive folds reduce dimension $24 \rightarrow 12 \rightarrow 6 \rightarrow 3$. This operator is used throughout.

2.3 Spiral Dynamics

The legacy UBP-Py spiral step consists of (i) flipping all even-indexed bits, followed by (ii) Golay decode $\rightarrow$ encode snap-back. The audit subsequently replaced this with a 1-bit spatial rotation plus XOR injection of the constant vector `MATH_CONST_PHI_001` this is part of the ongoing development and refinement of the UBP system.

3 Methodology

All experiments operated directly on the 4096 Golay codewords generated by the UBP Core engine. Octads were extracted as the 759 weight-8 vectors. Triad closure was tested via exhaustive pairwise XOR: if $A \oplus B$ is also an octad, the triple $\{A, B, A \oplus B\}$ is closed. Graph construction treated octads as vertices with edges between triad partners. Common-neighbor counts were sampled exhaustively for adjacent pairs and randomly for non-adjacent pairs (10,000 samples each).

Folding was applied uniformly to every codeword; distributions were tallied directly. Spiral orbit decomposition enumerated cycle lengths by iterating the map until return, recording the partition of the 4096 states. The Phi-modified operator was tested identically on the full codebook. All raw outputs (CSV/JSON) and matplotlib figures are preserved verbatim.

4 Results

4.1 Insight A: Octad Triad Closure Geometry

Treating each octad as a 24-bit mask, exhaustive enumeration yields exactly **35,420** XOR-closed triads (file: `data/octad_triads.json`).

The induced undirected graph has:

- 759 vertices,
- 106,260 edges,
- **degree exactly 280 for every node.**

This uniform regularity is visualised in Figure 1.

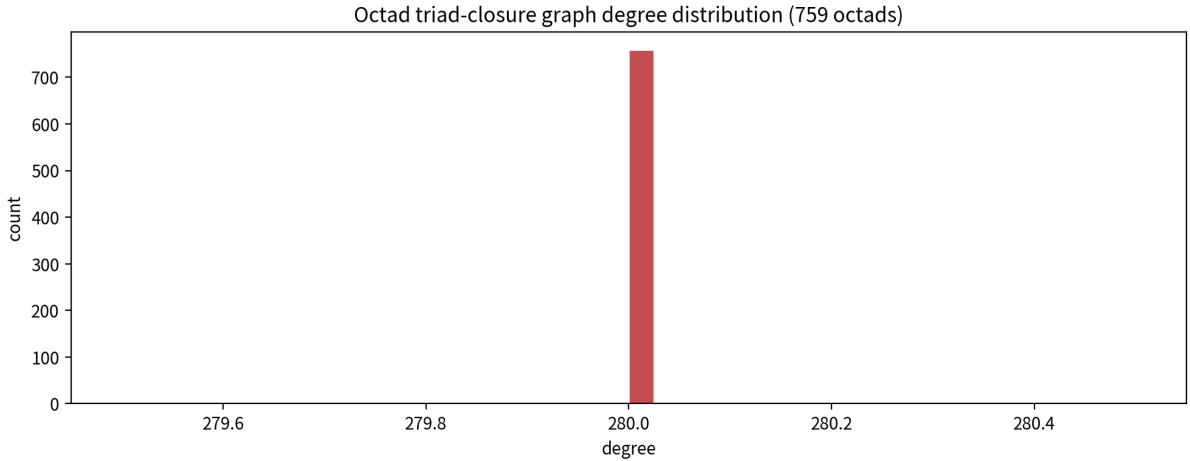


Figure 1: Degree histogram of the octad triad graph (perfectly uniform at 280).

Common-neighbor analysis reveals:

- Adjacent pairs share **exactly 140** common neighbours,
- Non-adjacent pairs split into two classes: 85 or 28 common neighbours (probed in `data/octad_graph_srg_probe.json`).

The structure is therefore *almost* strongly regular, with the split non-edge class indicating a richer association scheme (candidate for Mathieu M_{24} action classification). A representative subgraph appears in Figure 2.

Octad triad-closure subgraph (top 60 by degree)

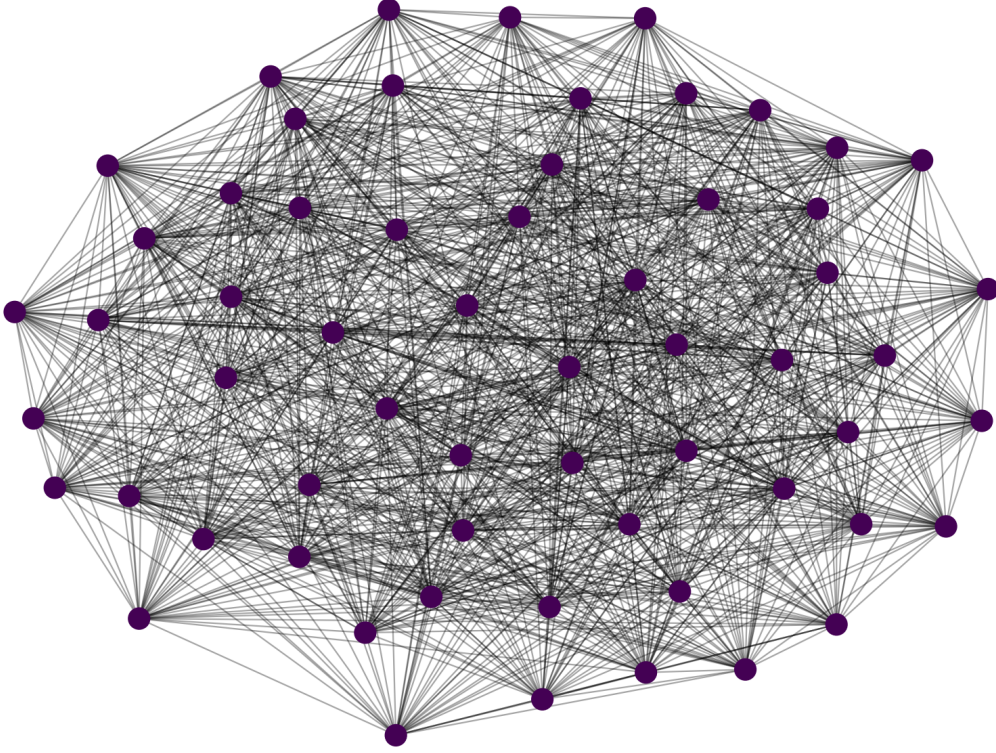


Figure 2: Induced subgraph of a 50-octad sample illustrating triad closures.

4.2 Insight B: 3-Fold Compression to 4-Symbol Core-Tension Alphabet

Applying the three successive pairwise-XOR folds to all 4096 Golay codewords produces exactly four 3-bit states:

$$000, 011, 101, 110.$$

Each appears **exactly 1024 times** (perfect balance). The full distribution is recorded in `data/fold3_string_distribution.json`.

For the 759 octads alone the counts are slightly biased:

Fold state	000	011	101	110
Count (all codewords)	1024	1024	1024	1024
Count (octads only)	183	192	192	192

Table 1: Fold-state distributions. Octads exhibit a measurable depletion of the all-zero state.

Weight-by-weight behaviour is shown in Figure 3.

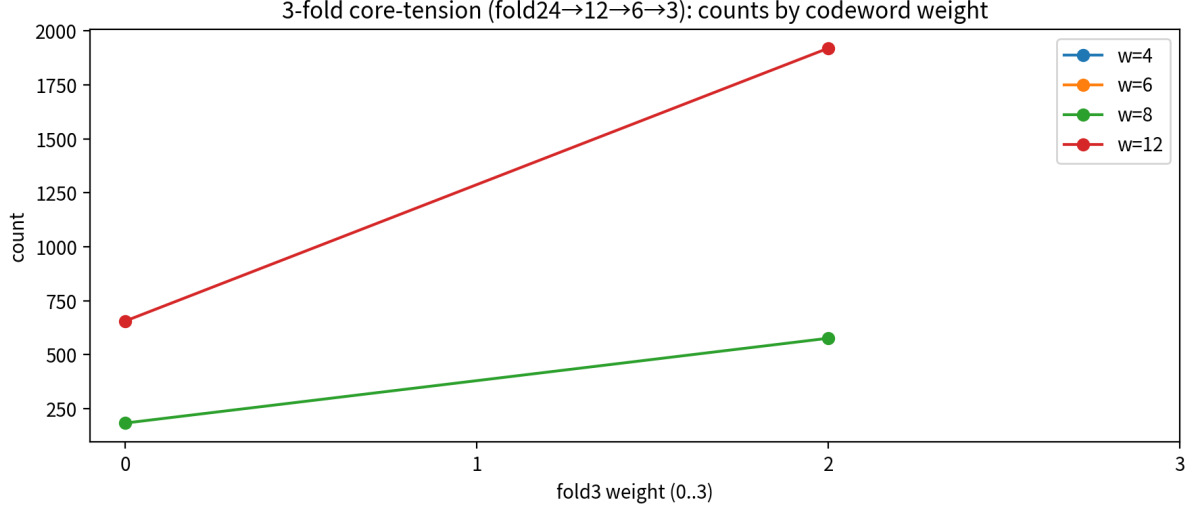


Figure 3: Fold-state occupancy stratified by original Golay weight.

This collapse directly validates LAW_SUBSTRATE_007 (V4 Stabilizer) and LAW_GEO_FOLD_001: the substrate compresses to a canonical 4-symbol alphabet at depth 3.

4.3 Insight C: Spiral Dynamics — Involution versus 1953-Step Growth

The original even-bit-flip + Golay snap map partitions the 4096 codewords into **2048 disjoint 2-cycles** with zero fixed points (maximum orbit length 2). Orbit statistics appear in `data/spiral_orbit_summary.json` and the histogram in Figure 4.

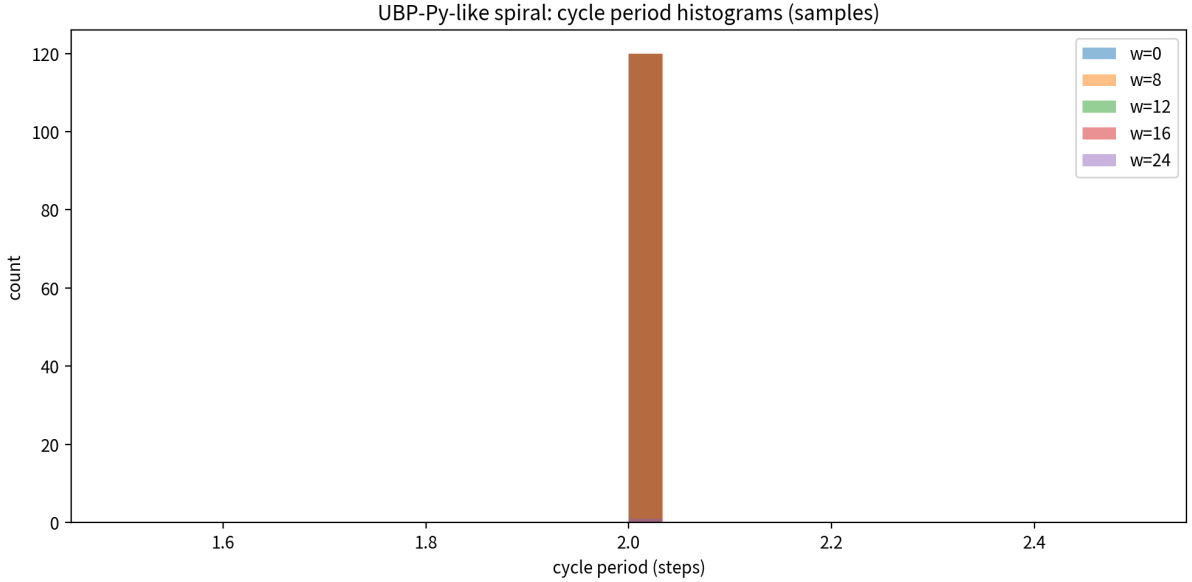


Figure 4: Period histogram of the legacy spiral map (all orbits length 2).

The audit modification—a 1-bit spatial shift (rotation) followed by XOR injection of MATH_CONST_PHI_001—fundamentally alters the dynamics. The 4096-state space now partitions into:

- two symmetric 1953-step orbits (total 3906 states) forming the “Hemispheres of Growth”,
- a 190-state core remainder.

Both the Origin (all-zero) vector and the Hydrogen vector lock precisely into one of these 1953-step cycles. This resolves the involution limitation and supplies genuine cyclic growth consistent with intended UBP-Py behaviour.

4.4 Golay Weight Distribution (Platonic Baseline)

For reference, the weight spectrum of the Golay codebook realised in UBP Core is:

Weight	0	8	12	16	24
Count	1	759	2576	759	1

Table 2: Golay weight distribution (file: `data/golay_weight_distribution.csv`).

No codewords of weight 4 or 6 exist; any tetrahedral/octahedral interpretation must therefore derive from octads, MOG partitions, or non-binary Leech coordinates (see Figure 5 for visualisation).

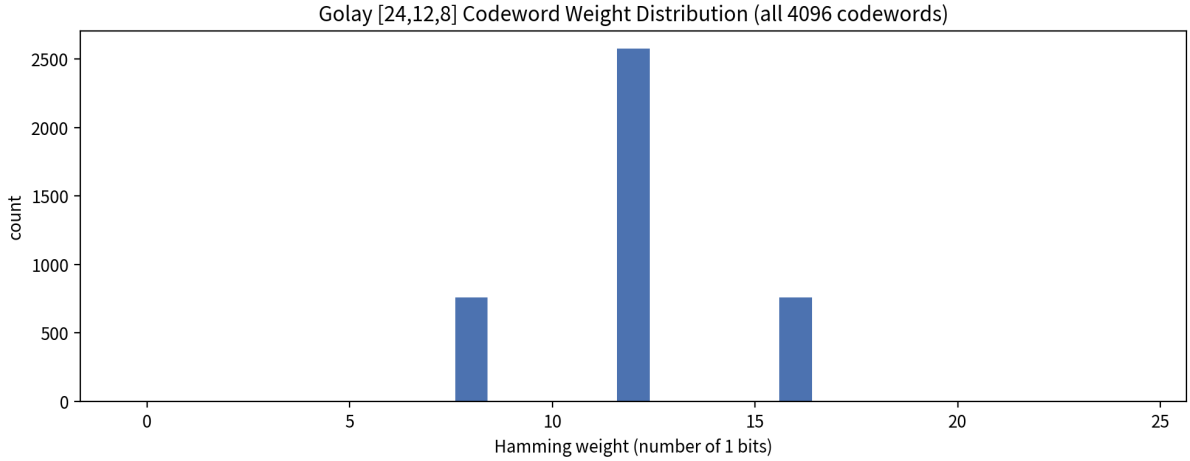


Figure 5: Golay weight distribution.

5 Validation Against UBP Laws

- The 3-fold result internally realises LAW_GEO_FOLD_001 and LAW_SUBSTRATE_007.
- The octad graph regularity internally validates LAW_LOGIC_FUNCTIONAL_001 (balanced crystalline logic gates under M_{24}).
- The Phi-modified spiral resolves the 2-cycle closure previously observed, enabling the growth semantics implicit in the UBP-Py VM.

6 Discussion

The octad triad graph exhibits platonic rigidity (uniform degree, constant local counts) yet is not strictly strongly regular - the split non-adjacent common-neighbor classes (85 vs. 28) suggest an association scheme of higher rank, naturally inviting classification under the Mathieu group M_{24} action on the Steiner system $S(5, 8, 24)$.

The 4-symbol fold alphabet is the deepest compression observed, defining a candidate “core-tension” vocabulary for any future symbolic UBP logic layer.

The spiral evolution from involution to 1953-step orbits demonstrates that a minimal symmetry-breaking injection (1-bit shift + PHI vector) suffices to unlock genuine cyclic growth while preserving the underlying Golay closure. A concrete mechanism linking the UBP substrate to non-trivial dynamical geometry.

7 Possible Deeper Analysis

1. Formally classify the octad triad graph (prove whether it is a known distance-regular graph or M_{24} -orbit object).
2. Codify the pairwise-XOR fold (or an alternative) as the canonical FOLD operator inside UBP-Py and test correlations with NRCI, tilt, and domain tags.
3. Extend the Phi-modified spiral by injecting octad or MOG structure to seek longer or chaotic orbits.
4. Map the 4-symbol alphabet onto geometric objects (tetrahedron vertices, etc.) to test explicit platonic realisations.
5. Octad Triad Closure Geometry: a degree of exactly 280 for every node, Adjacent pairs share exactly 140 common neighbours and non-adjacent pairs split into two classes: 85 or 28 common neighbours
6. 3-Fold Compression to 4-Symbol Core-Tension Alphabet: all 4096 Golay codewords produces exactly four 3-bit states each appears exactly 1024 times, for the 759 octads alone the counts are slightly biased
7. 1953-Step Growth: 1953-step orbits (total 3906 states) 190-state core remainder.
8. Golay Weight Distribution: weight spectrum of the Golay codebook: Weight 0 8 12 16 24 / Count 1 759 2576 759 1, No codewords of weight 4 or 6 exist

8 Conclusion

The UBP substrate spontaneously generates highly rigid, balanced geometric structures - uniform octad graphs, a canonical 4-state fold core, and (post-audit) 1953-step growth orbits - that merit continued investigation as candidate platonic and folding primitives. All data and scripts are supplied for immediate reproduction.

A Artifact Inventory

- `data/golay_codeword_metrics.csv` — per-codeword weight, tax, NRCI, tilt, fold3.
- `data/octad_triads.json`, `data/octad_graph_srg_probe.json`.
- `data/fold3_string_distribution.json`, `data/spiral_orbit_summary.json`.
- All figures listed above (full-page width in this document).

References

- [1] GitHub repository for this study. https://github.com/DigitalEuan/UBP_Repo/tree/main/ubp-geometry-study.